

Tendai NR Gwanzura

PHD CANDIDATE

Florida International University

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I am a highly motivated and accomplished epidemiologist, data scientist and early career researcher with experience in academic and government settings seeking to leverage my experience in government and academic institutions.

Education

Florida International University

PHD IN PUBLIC HEALTH EPIDEMIOLOGY

Miami, United States

2021-

Claremont Graduate University

MASTERS IN PUBLIC HEALTH

California, United States

2020

University of Zimbabwe

BACHELOR OF MEDICAL AND LABORATORY SCIENCE HONORS

Harare, Zimbabwe

2014

Professional Experience

Individual National Research Service Award Fellow

Florida International
University

NATIONAL INSTITUTES OF HEALTH (NIH)

May 2023 - May 2025

Demonstrated exceptional leadership and advanced research acumen in conducting independent research with Florida Department of Health data. Employed innovative methods, including multilevel survival analysis, to analyze over 5 million COVID-19 cases and 120,000 HIV diagnoses as part of Aim 1, showcasing a deep commitment to cutting-edge methodologies.

- Led the project, funded by the highly prestigious and competitive NIH F31 grant valued at 96,000 dollars over two years. This grant underscores the recognition of superior research skills and leadership abilities, further emphasizing the project's significance and impact.
- Showcased groundbreaking Aim 1 findings on COVID-19 mortality disparities in individuals with HIV pre- and post-vaccination at the FIU research seminar. Submitted research to the Society of Epidemiologic Research conference and a 2024 journal publication, affirming the project's rigor and contribution to the scientific community.

Graduate Assistant

Florida International
University

FLORIDA INTERNATIONAL UNIVERSITY

August 2021 - August 2025

Collaborated seamlessly and adaptably with distinguished Epidemiology faculty on multiple NIH-funded R01 and K award projects worth over 2 million dollars, focusing on gender-specific care within the Ryan White program.

- Partnered on impactful studies using ecological momentary assessment tools, to evaluate medication adherence using Wisepill or self-reported measures and the feasibility of an mHealth care intervention. Contributed to two published studies examining self-reported adherence based on diverse demographic and psychosocial factors.
- Contributed significantly to research enhancing women's involvement in opioid use disorder care through a novel recovery navigation system led by women in recovery, with a pending paper under review. Applied R, SAS, and GIS for comprehensive analysis, showcasing proactive and innovative methods in a dynamic research setting.
- Led survey creation for 1000 participants, proficiently cleaning and analyzing data with R, SAS, and Python to investigate stigma, provider-patient dynamics, and psychosocial factors' impact on ART adherence and viral suppression.
- Successfully contributed to the publication of three projects using methods such as Bayesian networks, showcasing a collaborative and dynamic approach.

Data Science Fellow: Women

DS4A

CORRELATION ONE

June 2022 - August 2022

Chosen for an exclusive 7-week data science Fellowship with a 2 percent acceptance rate featuring hands-on learning with real-world business cases.

- Led a six-member team to create a predictive model for variables associated with heightened pesticide levels in individuals (2009-2020) using NHANES data. Demonstrated proficiency in diverse tools, employing advanced regression and predictive models with Python (Matplotlib, Seaborn, Panda) and R (Tidymodels, Lasso Regression).
- Presented the team's findings to peers and business recruiters, earning an excellent grade for the project's depth and quality. Successfully navigated the Fellowship's challenging curriculum, highlighting adaptability, leadership, and advanced data science skills.

Data Science Fellow: Empowerment

DS4A

CORRELATION ONE

October 2021 - December 2022

Selected for an exclusive 14-week data science Fellowship with a highly competitive 2 percent acceptance rate, offering immersive learning with real-world business cases. Extensively trained in advanced Python methods during the Fellowship, applying cutting-edge techniques, including neural networks using TensorFlow to enhance my machine learning model's capabilities.

- Led a team of three to develop a state-of-the-art machine learning model, achieving an exceptional 98 percent accuracy in detecting brain tumors through MRI image classification. Presented dynamic results on a live dashboard powered by the Anvil app, showcasing real-time outputs throughout the program.
- As a DS4A ambassador, successfully recruited students from underrepresented groups into the Fellowship program, contributing to a diverse and dynamic learning environment. Placed two students at esteemed companies like Target, underscoring leadership and commitment to fostering inclusivity in the data science realm.

Health Program Analyst

California

LOS ANGELES COUNTY DEPARTMENT OF PUBLIC HEALTH

November 2020 - July 2021

Generated 200+ line lists by extracting crucial data from outbreak investigator notes on vaccination cases and demographic factors. Designed a daily dashboard to convey essential information to clinicians and stakeholders, emphasizing proficiency in data analysis and a results-driven approach to support effective health program strategies.

- Selected for the epidemiological curve team based on exceptional performance in an exam, contributing to the creation of daily epidemiologic curves for presentation at diverse homeless shelters. Generated comprehensive data reports for the outbreak, utilizing biostatistical software such as SAS.
- Collaborated seamlessly with Outbreak Investigators, addressing data and epidemiology needs for investigating site outbreaks affecting individuals experiencing homelessness and service planning in Los Angeles. Spearheaded communication with outbreak sites, overseeing line list follow-ups of cases and contacts, including daily monitoring, to achieve measurable outcomes.

Data Analyst intern

Claremont Graduate University

COMMUNITY TRANSLATIONAL RESEARCH INSTITUTE

February 2020 - June 2020

Enhanced data collection processes for the Diabetes Prevention Program through vigilant monitoring and development. Conducted thorough data cleaning in the electronic tracking system, delivering weekly reports with 100

- Applied advanced GIS techniques in ArcGIS to identify and analyze hotspots and activity spaces in Pomona Valley linked to high diabetes incidence. Utilized SAS and SPSS for statistical analyses, effectively managing a dataset of over 500 people to scrutinize participant demographics and program outcomes.
- Collaborated with research staff to contribute to research papers on the Stop Diabetes in its Tracks program outcomes. Presented compelling findings at the master's capstone thesis, impressing professors and earning an excellent grade, demonstrating effective communication and analytical prowess.

Embryologist intern

Bulawayo

BULAWAYO ASSISTED REPRODUCTIVE TECHNOLOGY CENTRE

January 2017 - July 2018

Performed semen analysis including sperm washing and freezing for over 50 patients. Aided in 6 intra uterine inseminations.

- Led the quality control team of the clinic and participated in invitro fertilization procedures that entailed ova extraction and culture.

Medical Laboratory Scientist

Bulawayo Harare

CIMAS MEDICAL LABS

December 2014 - April 2018

Executed a comprehensive array of chemical, microbiologic, immunologic, and hematological laboratory procedures as a proficient Medical Laboratory Scientist. Expertly operated and maintained laboratory equipment, adeptly troubleshooting and resolving instrument issues, ensuring smooth and accurate testing processes.

- Demonstrated efficiency in handling over 120 patient samples daily, including skillful phlebotomy, resulting in a daily collection of 10 samples.
- Spearheaded monthly and annual internal audits to secure ISO accreditation, showcasing meticulous attention to quality control and compliance standards.
- Achieved a remarkable 50
- Drove community health education initiatives, exhibiting a commitment to improving patient health outcomes.

Computer Skills

1. GIS Products {ArcGIS Pro, Online, Desktop, Enterprise, Server, and App Development using Web App Builder and other software, Insights, Spatial Analysis,QGIS,KNIME}
2. Advanced Statistical Analyses {Python, SAS,R, STATA, SPSS, SQL, MPlus, Tableau}
3. Reproducible Report {Quarto, Markdown/Rmarkdown, Jupyter Notebook}
4. DevOps {Git, GitHub}
5. Microsoft {Excel, Word, Power Point and more}

Academic Publications

1. Trepka, M.J., Ward, M.K., Fennie, K., Sheehan, D.M., Fernandez, S.B., Li, T., Jean-Gilles, M., Dvieux, J.G., Ibaez, G.E., Gwanzura, T. and Nawfal, E., 2023. Patient–Provider Relationships and Antiretroviral Therapy Adherence and Durable Viral Suppression Among Women with HIV, Miami-Dade County, Florida, 2021–2022. *AIDS patient care and STDs*, 37(7), pp.361-372.
2. Gray, A., Ward, M.K., Fernandez, S.B., Nawfal, E.S., Gwanzura, T., Li, T., Sheehan, D.M., Jean-Gilles, M., Beach, M.C., Ladner, R.A. and Trepka, M.J., 2023. Exploring the use of self-management strategies for antiretroviral therapy adherence among women with HIV in the Miami-Dade County Ryan White Program. *Journal of Behavioral Medicine*, pp.1-13.
3. Ward MK, Gwanzura T, Rojas RR, Trepka MJ, Bursac Z, Wagner EF. Nonfatal opioid-related overdoses treated by emergency medical services in Florida, before and during the COVID-19 pandemic. *Preventive Medicine Reports*. 2023;31:102102. doi:10.1016/j.pmedr.2022.102102
4. Chipidza W, Krewson C, Gatto N, Akbaripourdibazar E, Gwanzura T. Ideological variation in preferred content and source credibility on Reddit during the COVID-19 pandemic. *Big Data & Society*. 2022;9(1):20539517221076490. doi:10.1177/20539517221076486.
5. Mshigeni SK, Rittenhouse A, Gwanzura T, Arroyo R, Vaughn C. Understanding Demographics Characteristics of E-Cigarette Users in California and the Need for Tailored Interventions. *American Journal of Health Education*. 2021;52(4):185-193. doi:10.1080/19325037.2021.1930610
6. Chipidza W, Akbaripourdibazar E, Gwanzura T, Gatto NM. Topic Analysis of Traditional and Social Media News Coverage of the Early COVID-19 Pandemic and Implications for Public Health Communication. *Disaster Medicine and Public Health Preparedness*. Published online March 3, 2021:1-8. doi:10.1017/dmp.2021.65.
7. Mtetwa S, Gwanzura TN. Interrogating Child Rights Teaching in Bindura Schools – Zimbabwe. *International Journal of Educational Sciences*. 2017;18(1-3):200-213. doi:10.1080/09751122.2017.1347233.

Extracurricular activity

Awards : Delta Omega Epsilon Chapter 2020 Inductee.

Grant : Ruth L. Kirschstein Predoctoral Individual National Research Service Award, National Institute on Minority Health, and Health Disparities worth 96 000 USD.

Volunteer work : Mentor at Sisters in Public Health : Engagement Committee member at Florida International University : National Chapter Chair at Sisters in Public Health

Hobbies and Interests: Tandem Skydiving, travelling, bungee jumping, reading, volunteer work, voice over productions.